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Culinary RAG API Spec

Freelance Project API Documentation

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Change Log

Version	Change	Date
1.0	Initial version.	May 6, 2026

API Request Overview

This API is for a Culinary RAG chat application. The frontend sends chat questions to the backend. The backend retrieves relevant culinary source chunks, generates an AI answer, stores the chat history, and stores the source citations used for the answer.

The API mainly expects requests for the following resources:

Resource	What the frontend/backend uses it for	Main Request Types
Documents	Store metadata about uploaded culinary books, PDFs, or source files.	Create, get, list, delete
Chunks	Store searchable text pieces from documents with vector embeddings.	List, search
Chats	Store each chat session shown in the sidebar.	Create, get, list, edit, delete
Messages	Store user and assistant messages inside a chat.	Create, list, get, delete
Message Sources	Store retrieved source chunks used by an assistant answer.	List sources for a message
Answer	Main RAG endpoint that accepts a user question and returns an AI answer with sources.	Ask question

Expected Request Types

User Action / System Action	Request	Purpose
User opens the app	GET /chats	Load chat history for the sidebar.
User clicks an existing chat	GET /chats/:chat_id	Load the selected chat and its messages.
User starts a new chat	POST /chats	Create a new empty chat session.
User renames a chat	PATCH /chats/:chat_id	Update the chat title.
User deletes a chat	DELETE /chats/:chat_id	Delete the chat and its messages.
User submits a question	POST /answer	Generate a RAG answer, save messages, and save sources.
User views source citations	GET /messages/:message_id/sources	Load source chunks for an assistant message.
Admin uploads/registers a document	POST /documents	Save document metadata.
Admin views documents	GET /documents	List available culinary source documents.
Admin deletes a document	DELETE /documents/:document_id	Remove a document and its chunks.

Backend/data pipeline lists chunks	GET /documents/:document_id/chunks	View chunks belonging to a document.
Backend searches knowledge base	POST /chunks/search	Retrieve relevant chunks using semantic search.

Main Frontend Flow

1. User opens the app
-> GET /chats
2. User starts a new chat
-> POST /chats
3. User asks a question
-> POST /answer
4. Backend saves:
 - user message
 - assistant message
 - retrieved source chunks
5. Frontend displays:
 - user question
 - AI answer
 - source citations
6. User returns later
-> GET /chats
-> GET /chats/:chat_id

Main Request Body Patterns

Create Chat Request

```
{
  "title": "Braising Questions"
}
```

Ask Question Request

```
{
  "chat_id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
  "question": "Why do chefs sear meat before braising?",
  "top_k": 5,
  "strictness": "balanced",
  "response_language": "english"
}
```

Create Document Request

```
{
  "title": "The Professional Chef",
  "total_page": 1230,
  "author": "The Culinary Institute of America",
  "published_at": "2011-09-13T00:00:00Z",
  "file_path": "uploads/the_professional_chef.pdf"
}
```

Main Response Body Patterns

Single Resource Response

```
{
  "id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
  "title": "Braising Questions",
  "created_at": "2026-05-06T07:10:00Z",
  "self": "https://culinary-rag-api.example.com/chats/6a6f5f3c-4c79-46a5-9a54-3333abcd4444"
}
```

Paginated List Response

```
{
  "entries": [
    {
      "id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
      "title": "Braising Questions",
      "created_at": "2026-05-06T07:10:00Z",
      "self": "https://culinary-rag-api.example.com/chats/6a6f5f3c-4c79-46a5-9a54-3333abcd4444"
    }
  ],
  "next": "https://culinary-rag-api.example.com/chats?offset=20&limit=20"
}
```

Error Response

```
{
  "Error": "No chat with this chat_id exists"
}
```

Create a Chat

Allows you to create a new chat session. If an initial question is provided, the API may generate a chat title from that question.

POST /chats

Request

Path Parameters

None

Request Body

Optional

Request Body Format

JSON

Request JSON Attributes

Name	Description	Required?
question	the LLM will generate a title based on the question.	No

Request Body Example

```
{
  "question": "How do I make delicious ice-cream"
}
```

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	201 Created	Chat was created successfully.

Response Examples

Success

```
Status: 201 Created
{
  "id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
  "title": "Process of making an icecream",
  "created_at": "2026-05-06T07:10:00Z",
  "self": "https://culinary-rag-api.example.com/chats/6a6f5f3c-4c79-46a5-9a54-3333abcd4444"
}
```

Get a Chat

Allows you to get an existing chat and its messages.

```
GET /chats/:chat_id
```

Request

Path Parameters

Name	Description
chat_id	ID of the chat.

Request Body

None

Request Body Format

None

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	200 OK	Chat exists.
Failure	404 Not Found	No chat with this chat_id exists.

Response Examples

Success

```
Status: 200 OK
{
  "id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
  "title": "Braising Questions",
  "created_at": "2026-05-06T07:10:00Z",
  "messages": [
    {
      "id": "7c9a9220-4d54-4a7e-a0f0-5555abcd6666",
      "role": "user",
      "content": "Why do chefs sear meat before braising?",
      "created_at": "2026-05-06T07:11:00Z"
    },
    {
      "id": "9f3171f0-0a24-4c56-bd12-7777abcd8888",
      "role": "assistant",
      "content": "Chefs sear meat before braising to build deeper flavor through browning.",
      "created_at": "2026-05-06T07:11:05Z",
      "sources": [
        {
          "id": 1,
          "document_title": "The Professional Chef",
          "page_start": 45,
          "page_end": 46,
          "similarity": 0.88,

```

```
      "chunk_text": "Braising often begins by browning the main item..."
    }
  ]
},
{
  "self": "https://culinary-rag-api.example.com/chats/6a6f5f3c-4c79-46a5-9a54-3333abcd4444"
}
```

Failure

```
Status: 404 Not Found
{
  "Error": "No chat with this chat_id exists"
}
```

List all Chats

Lists all chat sessions using pagination.

```
GET /chats
GET /chats?offset=0&limit=20
```

Request

Path Parameters

None

Query Parameters

Name	Description	Required?
offset	Number of chats to skip. Default is 0.	No
limit	Maximum number of chats to return. Default is 20.	No

Request Body

None

Request Body Format

None

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	200 OK	Chats were returned successfully.

Response Notes

- Each chat in the response includes a self property whose value is the complete URL for that chat.
- The entries property is an array with the chats for this page.
- The next property is a URL with query parameters offset and limit set to the next page values.

Response Examples

Success

```
Status: 200 OK
{
  "entries": [
    {
      "id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
      "title": "Braising Questions",
      "created_at": "2026-05-06T07:10:00Z",
      "self": "https://culinary-rag-api.example.com/chats/6a6f5f3c-4c79-46a5-9a54-3333abcd4444"
    }
  ],
  "next": "https://culinary-rag-api.example.com/chats?offset=20&limit=20"
}
```


Delete a Chat

Allows you to delete a chat. Deleting a chat also deletes all messages and sources connected to that chat.

```
DELETE /chats/:chat_id
```

Request

Path Parameters

Name	Description
chat_id	ID of the chat.

Request Body

None

Request Body Format

None

Response

Response Body Format

Success: No body

Failure: JSON

Response Statuses

Outcome	Status Code	Notes
Success	204 No Content	Chat was deleted successfully.
Failure	404 Not Found	No chat with this chat_id exists.

Response Examples

Success

```
Status: 204 No Content
```

Failure

```
Status: 404 Not Found
{
  "Error": "No chat with this chat_id exists"
}
```

Ask a Question

Allows the user to ask a culinary question. The API retrieves relevant chunks, generates an AI answer, saves the user message, saves the assistant message, and saves the sources used for the answer.

POST /answer

Request

Path Parameters

None

Request Body

Required

Request Body Format

JSON

Request JSON Attributes

Name	Description	Required?
chat_id	ID of the chat. If not provided, the API may create a new chat.	No
question	The user's culinary question.	Yes
top_k	Number of source chunks to retrieve. Default is 5.	No
strictness	Answer mode. Possible values are strict, balanced, creative, or recipe.	No
response_language	Response language. Possible values are auto, english, or japanese.	No

Request Body Example

```
{
  "chat_id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
  "question": "Why do chefs sear meat before braising?",
  "top_k": 5,
  "strictness": "balanced",
  "response_language": "english"
}
```

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	200 OK	Answer was generated successfully.
Failure	400 Bad Request	The request body is missing question.
Failure	404 Not Found	No chat with this chat_id exists.
Failure	500 Internal Server Error	The answer could not be generated.

Response Examples

Success

```
Status: 200 OK
{
  "chat_id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
  "user_message": {
    "id": "7c9a9220-4d54-4a7e-a0f0-5555abcd6666",
    "role": "user",
    "content": "Why do chefs sear meat before braising?",
    "created_at": "2026-05-06T07:20:00Z"
  },
  "assistant_message": {
    "id": "9f3171f0-0a24-4c56-bd12-7777abcd8888",
    "role": "assistant",
    "content": "Chefs sear meat before braising to build deeper flavor through browning. The browned surface adds complexity to the final sauce and improves the overall taste of the dish.",
    "created_at": "2026-05-06T07:20:05Z"
  },
  "sources": [
    {
      "id": 1,
      "document_title": "The Professional Chef",
      "page_start": 45,
      "page_end": 46,
      "similarity": 0.88,
      "chunk_text": "Braising often begins by browning the main item..."
    }
  ]
}
```

Failure

```
Status: 400 Bad Request
{
  "Error": "The request body is missing question"
}
```

```
Status: 404 Not Found
{
  "Error": "No chat with this chat_id exists"
}
```

```
Status: 500 Internal Server Error
{
  "Error": "The answer could not be generated"
}
```

Create a Message

Allows you to create a new message in a chat.

POST /chats/:chat_id/messages

Request

Path Parameters

Name	Description
chat_id	ID of the chat.

Request Body

Required

Request Body Format

JSON

Request JSON Attributes

Name	Description	Required?
role	Role of the message. Possible values are user, assistant, or system.	Yes
content	Text content of the message.	Yes

Request Body Example

```
{
  "role": "user",
  "content": "How do I make a good stock?"
}
```

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	201 Created	Message was created successfully.
Failure	400 Bad Request	The request body is missing role or content.
Failure	404 Not Found	No chat with this chat_id exists.

Response Examples

Success

```
Status: 201 Created
{
  "id": "7c9a9220-4d54-4a7e-a0f0-5555abcd6666",
  "chat_id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
}
```

```
"role": "user",
"content": "How do I make a good stock?",
"created_at": "2026-05-06T07:15:00Z",
"self": "https://culinary-rag-api.example.com/messages/7c9a9220-4d54-4a7e-a0f0-5555abcd6666",
"chat": "https://culinary-rag-api.example.com/chats/6a6f5f3c-4c79-46a5-9a54-3333abcd4444"
}
```

Failure

Status: 400 Bad Request

```
{
  "Error": "The request body is missing at least one of the required attributes"
}
```

Status: 404 Not Found

```
{
  "Error": "No chat with this chat_id exists"
}
```

List all Messages for a Chat

Lists all messages for a chat.

GET /chats/:chat_id/messages

Request

Path Parameters

Name	Description
chat_id	ID of the chat.

Request Body

None

Request Body Format

None

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	200 OK	Messages were returned successfully.
Failure	404 Not Found	No chat with this chat_id exists.

Response Examples

Success

```
Status: 200 OK
[
  {
    "id": "7c9a9220-4d54-4a7e-a0f0-5555abcd6666",
    "chat_id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
    "role": "user",
    "content": "How do I make a good stock?",
    "created_at": "2026-05-06T07:15:00Z",
    "sources": [],
    "self": "https://culinary-rag-api.example.com/messages/7c9a9220-4d54-4a7e-a0f0-5555abcd6666"
  },
  {
    "id": "9f3171f0-0a24-4c56-bd12-7777abcd8888",
    "chat_id": "6a6f5f3c-4c79-46a5-9a54-3333abcd4444",
    "role": "assistant",
    "content": "A good stock starts with cold water, bones or vegetables, aromatics, and gentle simmering.",
    "created_at": "2026-05-06T07:15:05Z",
    "sources": [
      {
        "id": 1,
        "document_title": "The Professional Chef",
        "page_start": 120,
        "page_end": 122,
```

```
      "similarity": 0.91,  
      "chunk_text": "Stocks are flavorful liquids prepared by simmering bones..."  
    }  
  ],  
  "self": "https://culinary-rag-api.example.com/messages/9f3171f0-0a24-4c56-bd12-7777abcd8888"  
}
```

Failure

```
Status: 404 Not Found  
{  
  "Error": "No chat with this chat_id exists"  
}
```

Create a Document

Allows you to create a new document.

POST /documents

Request

Path Parameters

None

Request Body

Required

Request Body Format

JSON

Request JSON Attributes

Name	Description	Required?
title	Title of the document. Up to 255 characters.	Yes
total_page	Total number of pages in the document.	Yes
author	Author of the document. Up to 255 characters.	No
published_at	Date and time when the document was published.	No
file_path	File path or storage path of the document. Must be unique.	Yes

Request Body Example

```
{
  "title": "The Professional Chef",
  "total_page": 1230,
  "author": "The Culinary Institute of America",
  "published_at": "2011-09-13T00:00:00Z",
  "file_path": "uploads/the_professional_chef.pdf"
}
```

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	201 Created	Document was created successfully.
Failure	400 Bad Request	The request body is missing at least one required attribute.
Failure	409 Conflict	A document with this file_path already exists.

Response Examples

Success

Status: 201 Created

```
{
  "document_id": "1fcd7a5a-9a5c-4c02-bf3f-81a29a0a1111",
  "title": "The Professional Chef",
  "total_page": 1230,
  "author": "The Culinary Institute of America",
  "created_at": "2026-05-06T07:00:00Z",
  "published_at": "2011-09-13T00:00:00Z",
  "status": "pending",
  "file_path": "uploads/the_professional_chef.pdf",
  "self": "https://culinary-rag-api.example.com/documents/1fcd7a5a-9a5c-4c02-bf3f-81a29a0a1111"
}
```

Failure

Status: 400 Bad Request

```
{
  "Error": "The request body is missing at least one of the required attributes"
}
```

Status: 409 Conflict

```
{
  "Error": "A document with this file_path already exists"
}
```

Get a Document

Allows you to get an existing document.

```
GET /documents/:document_id
```

Request

Path Parameters

Name	Description
document_id	ID of the document.

Request Body

None

Request Body Format

None

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	200 OK	Document exists.
Failure	404 Not Found	No document with this document_id exists.

Response Examples

Success

```
Status: 200 OK
{
  "document_id": "1fcd7a5a-9a5c-4c02-bf3f-81a29a0a1111",
  "title": "The Professional Chef",
  "total_page": 1230,
  "author": "The Culinary Institute of America",
  "created_at": "2026-05-06T07:00:00Z",
  "published_at": "2011-09-13T00:00:00Z",
  "status": "processed",
  "file_path": "uploads/the_professional_chef.pdf",
  "self": "https://culinary-rag-api.example.com/documents/1fcd7a5a-9a5c-4c02-bf3f-81a29a0a1111"
}
```

Failure

```
Status: 404 Not Found
{
  "Error": "No document with this document_id exists"
}
```

List all Documents

Lists all documents using pagination.

```
GET /documents
GET /documents?offset=0&limit=10
```

Request

Path Parameters

None

Query Parameters

Name	Description	Required?
offset	Number of documents to skip. Default is 0.	No
limit	Maximum number of documents to return. Default is 10.	No

Request Body

None

Request Body Format

None

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	200 OK	Documents were returned successfully.

Response Examples

Success

```
Status: 200 OK
{
  "entries": [
    {
      "document_id": "1fcd7a5a-9a5c-4c02-bf3f-81a29a0a1111",
      "title": "The Professional Chef",
      "total_page": 1230,
      "author": "The Culinary Institute of America",
      "created_at": "2026-05-06T07:00:00Z",
      "published_at": "2011-09-13T00:00:00Z",
      "status": "processed",
      "file_path": "uploads/the_professional_chef.pdf",
      "self": "https://culinary-rag-api.example.com/documents/1fcd7a5a-9a5c-4c02-bf3f-81a29a0a1111"
    }
  ],
  "next": "https://culinary-rag-api.example.com/documents?offset=10&limit=10"
}
```

Delete a Document

Allows you to delete a document. Deleting a document also deletes all chunks that belong to the document.

```
DELETE /documents/:document_id
```

Request

Path Parameters

Name	Description
document_id	ID of the document.

Request Body

None

Request Body Format

None

Response

Response Body Format

Success: No body

Failure: JSON

Response Statuses

Outcome	Status Code	Notes
Success	204 No Content	Document was deleted successfully.
Failure	404 Not Found	No document with this document_id exists.

Response Examples

Success

```
Status: 204 No Content
```

Failure

```
Status: 404 Not Found
{
  "Error": "No document with this document_id exists"
}
```

Search Chunks

Allows you to search document chunks using semantic search.

POST /chunks/search

Request

Path Parameters

None

Request Body

Required

Request Body Format

JSON

Request JSON Attributes

Name	Description	Required?
query	Search query text.	Yes
top_k	Number of chunks to return. Default is 5.	No
max_distance	Maximum cosine distance allowed.	No

Request Body Example

```
{
  "query": "How do I properly braise meat?",
  "top_k": 5,
  "max_distance": 0.75
}
```

Response

Response Body Format

JSON

Response Statuses

Outcome	Status Code	Notes
Success	200 OK	Matching chunks were returned successfully.
Failure	400 Bad Request	The request body is missing query.

Response Examples

Success

```
Status: 200 OK
{
  "query": "How do I properly braise meat?",
  "entries": [
    {
      "chunk_id": "4d9f7e5d-2a88-41c0-b9a0-1111abcd2222",
      "document_id": "1fcd7a5a-9a5c-4c02-bf3f-81a29a0a1111",
      "document_title": "The Professional Chef",

```

```
"page_start": 45,  
"page_end": 46,  
"chunk_text": "Braising is a combination cooking method that uses both dry and moist heat.",  
"similarity": 0.87,  
"self": "https://culinary-rag-api.example.com/chunks/4d9f7e5d-2a88-41c0-b9a0-1111abcd2222"  
}  
]  
}
```

Failure

```
Status: 400 Bad Request  
{  
  "Error": "The request body is missing query"  
}
```